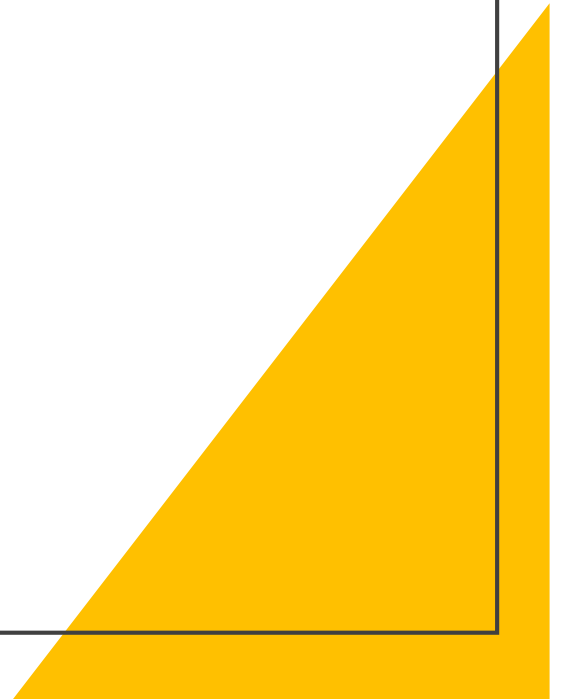


Data Analytics Technology Introduction

Catarina Silva

catarina@dei.uc.pt

2026



Agenda

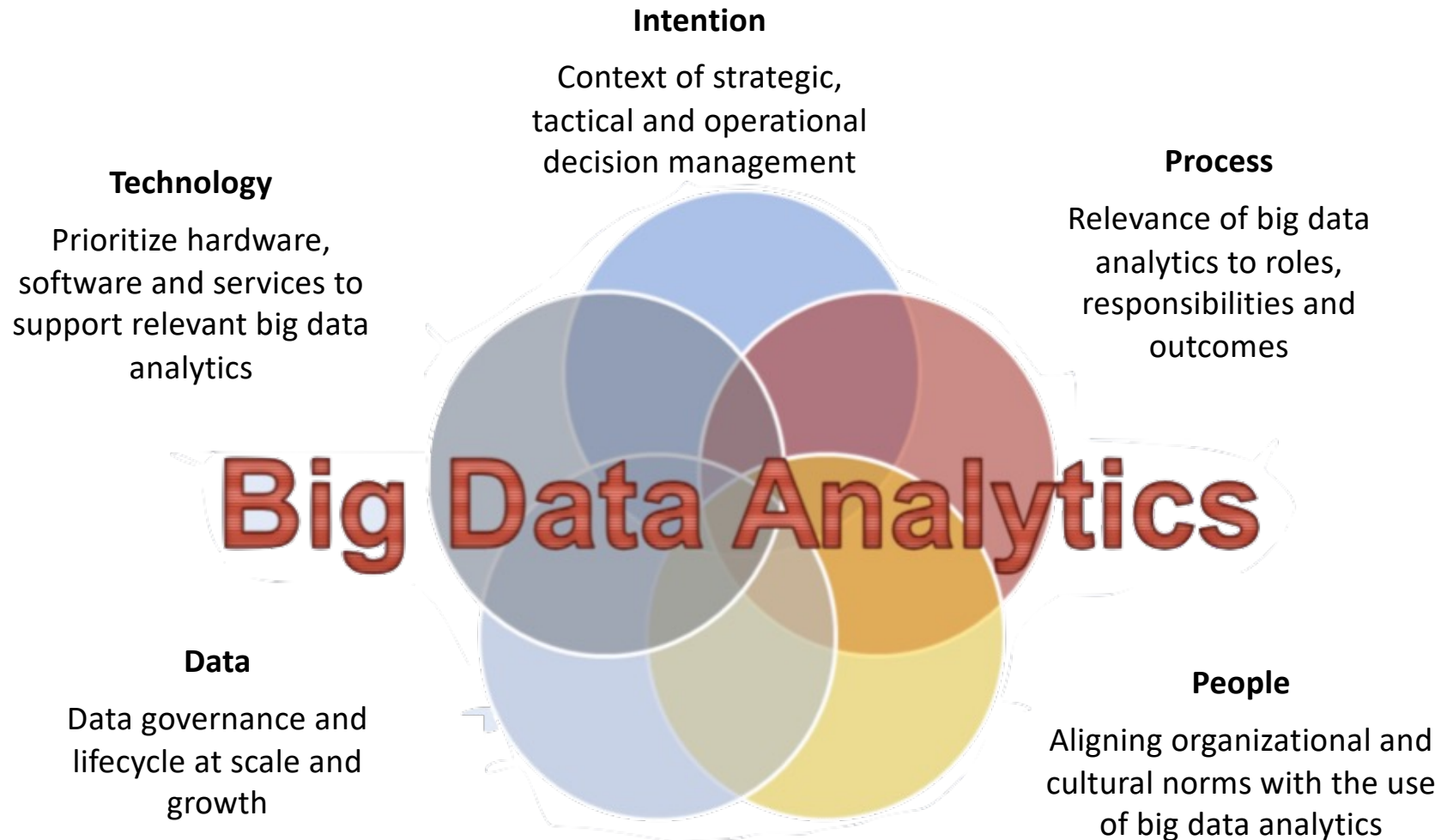
- Data analytics:
 - introduction and objectives
- Types of analysis
 - descriptive,
 - diagnostic,
 - predictive,
 - prescriptive,
 - decisive
- Data Analytics cycle

Data Analytics

- Examines (large) amounts of data to discover **patterns**, **relationships**, and other information
- It helps organizations take control of their data and use it to identify new **opportunities** that can lead to better business **decisions**, more **efficient** operations, higher **profits** and **happier** customers.

Analytics: Objectives

- **Cost reduction**
 - Big data analytics technologies bring significant cost advantages when it comes to storing large amounts of data as well as identifying more efficient ways of doing business.
- **Faster, better decision making**
 - The speed of storage and infrastructure, combined with the ability to analyze new data sources, means companies can analyze information immediately and make decisions based on what they have learned.
- **New products and services**
 - With the ability to assess customer needs and satisfaction through analytics comes the power to give customers what they want.



Types of Analytics

- **Descriptive:** To analyze and explore the dataset(s) to understand the data and analyze the performance of the business
- **Diagnostic:** To determine what happened and why
- **Predictive:** To analyze current and historical data to determine what is likely (not) to happen
- **Prescriptive:** To computationally develop and analyze alternatives that can be courses of action - tactical or strategic - that can discover the unexpected
- **Decisive:** To visualize information and recommend courses of action to facilitate human decision making when presented with a set of alternatives.

Descriptive Analysis: What happened?

- Identify attributes, then evaluate attributes
- Estimate the magnitude to correlate the relative contribution of each attribute to the final solution
- Accumulate more data instances from data sources
- If possible, perform the assessment, classification and categorization steps quickly
- Produces a measure of adaptability within the OODA loop (Observe-Orient-Decide-Act)

Diagnostic Analysis: Why did it happen?

- Get started with descriptive analysis
- Extract patterns from large amounts of data (through data mining)
- Correlate data types to explain short-term behavior - past and present
- Estimate linear / non-linear behavior not easily identifiable by other approaches.
- Example: ranking past insurance claims, estimating the number of future claims to be flagged for investigation with a high probability of being fraudulent

Predictive Analysis: What is likely to happen in the future?

- Start with descriptive and diagnostic analysis
- Choosing the correct data based on domain knowledge and relationships between variables
- Choosing the right techniques to gain insight into potential outcomes
- Determine the probability of possible outcomes, given the initial boundary conditions
- Example: based on data from the previous year's orders and the weather forecast, predict how many umbrellas will be sold in December.

Prescriptive Analysis: What is the best course of action?

- Get started with predictive analytics
- Determining what should happen and how to do it
- Determining mitigating factors leading to desirable/undesirable outcomes
- “What-if” analysis with local or global optimization
- Examples: (i) find the best set of prices and advertising frequency to maximize revenue; (ii) find the right set of businesses to achieve that goal

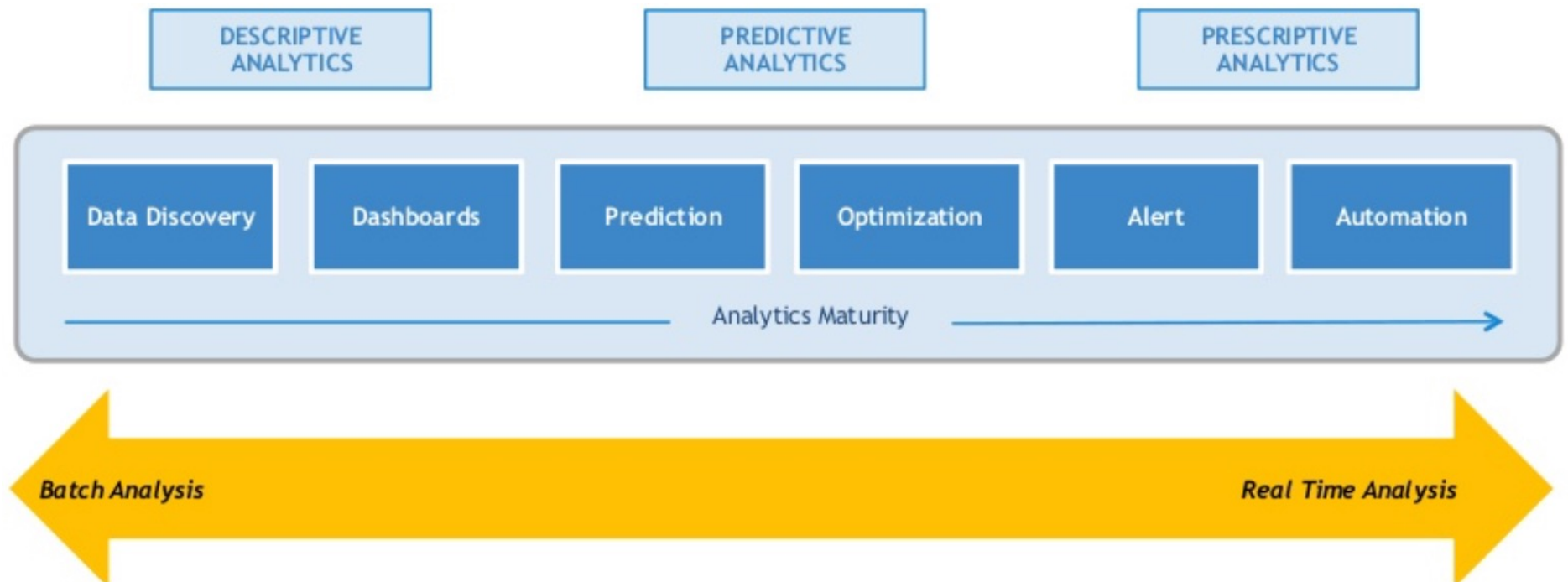
Decisive Analysis

- Given a set of decision alternatives, choose the following course of action (may not be ideal)
- View alternatives - whole or partial subset
- Conduct exploratory analysis - what-if and why
- How do I get there from here?
- How do I get here from there?

Classification of Types of Analysis

	Passive	Active
Deductive	Descriptive	Diagnostic
Inductive	Predictive	Prescriptive

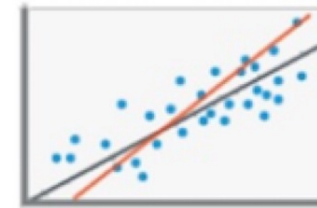
- **Inductive:** from the particular to the general
- **Deductive:** from the general to the particular



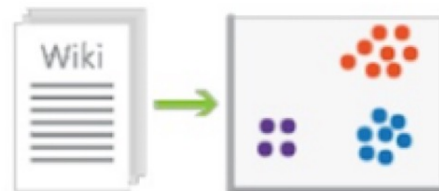
Types of problems



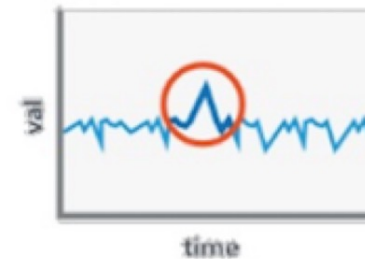
Classification
(supervised – predictive)



Regression
(supervised – predictive)

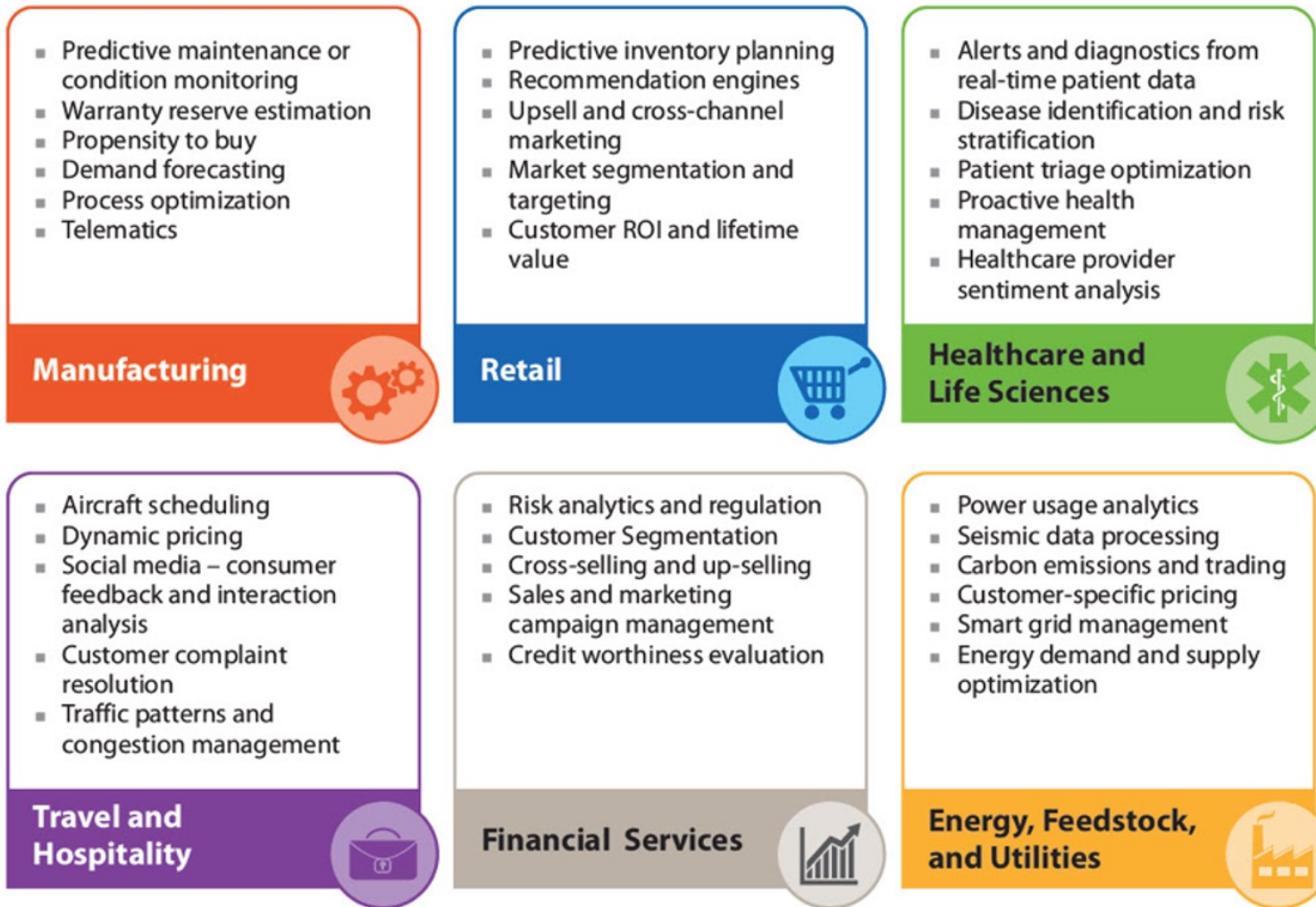


Clustering
(unsupervised – descriptive)

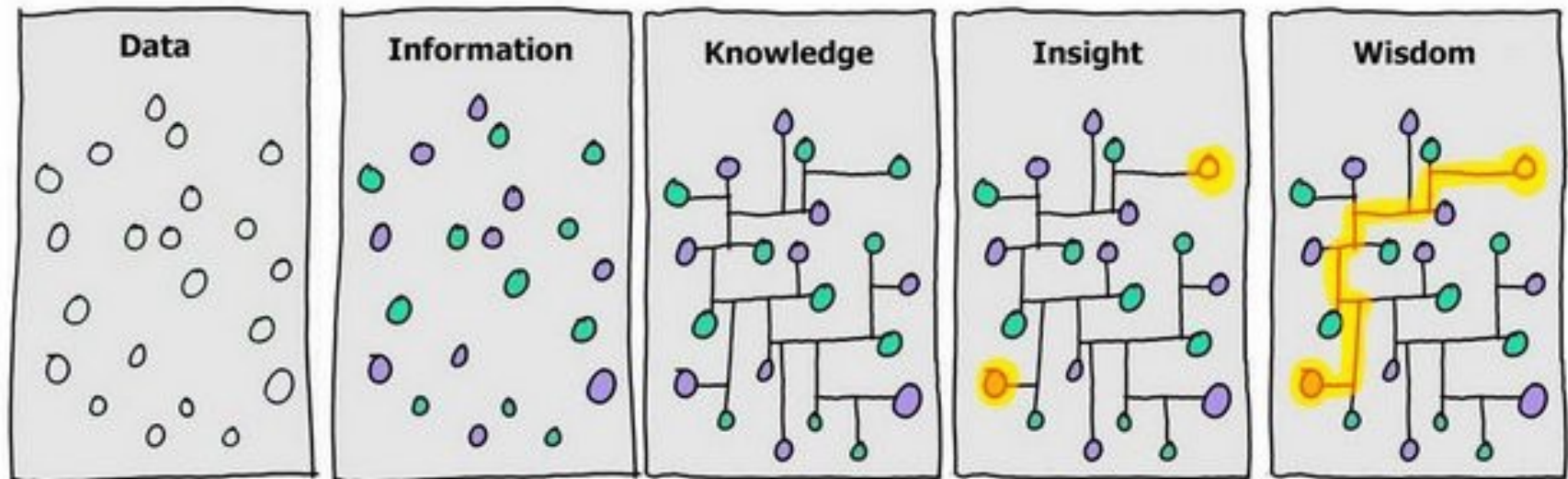


Anomaly Detection
(unsupervised – descriptive)

Applications



DIK(I)W Model



Source: towardsdatascience

DIKW Example

- **Data:** It is raining.
- **Information:** The temperature dropped 5 degrees, the humidity rose 5% in one hour and it started to rain at 15:00.
- **Knowledge:** A rapid increase in humidity, accompanied by a drop in temperature caused by areas of low pressure, is likely to make the atmosphere unable to retain moisture and rain.
- **Wisdom:** Based on observations and the mathematical model, we can predict why and when it will rain in the future, and we can do this so quickly and systematically that it won't require much analysis. We already have an understanding of all the interactions that take place between evaporation, air currents, temperature gradients, changes and rainfall.

DATA



SORTED



ARRANGED

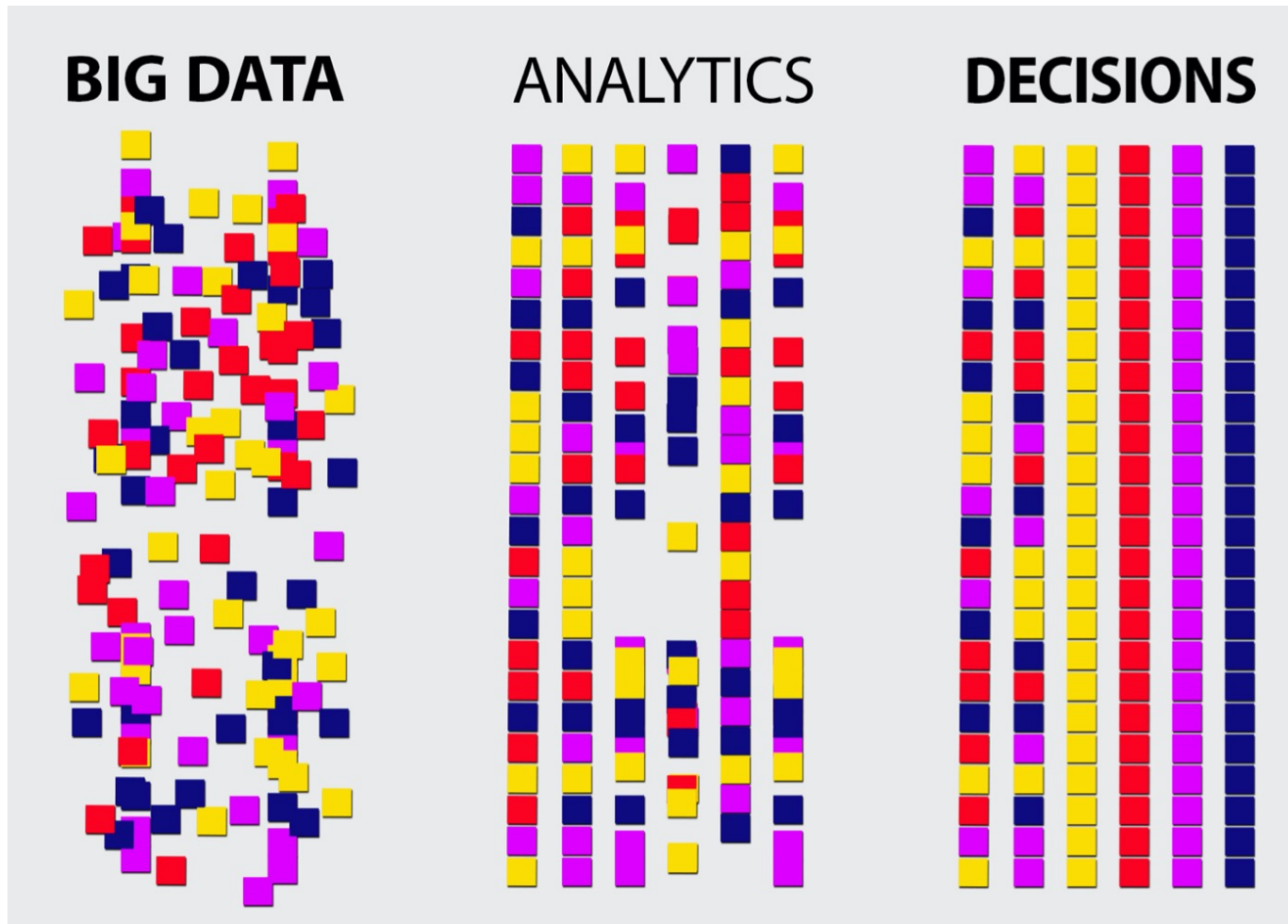


PRESENTED
VISUALLY



EXPLAINED
WITH A STORY



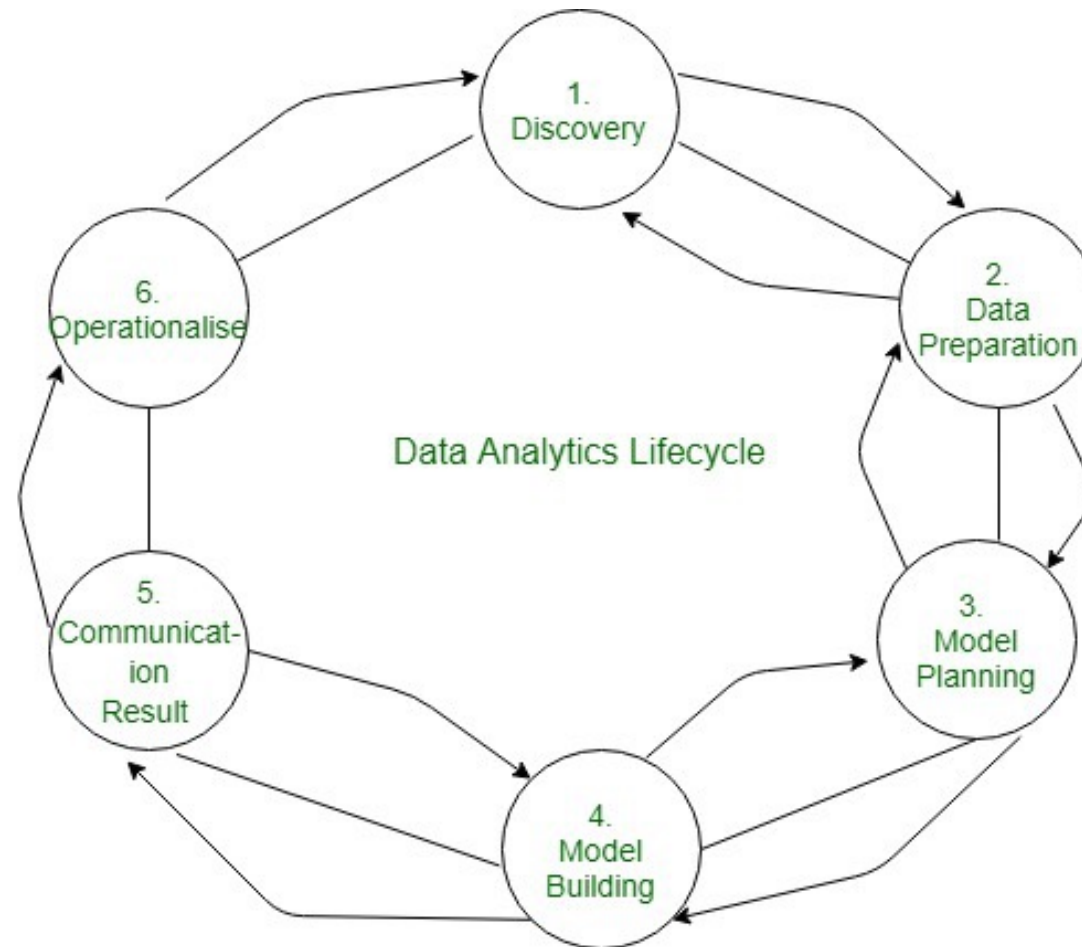


Source: Ahmed & Pathan, Data Analytics

Data Analytics Plan

- It's always difficult to know where to start a data analysis project:
 - Where to start, what are the project objectives?
 - How to get familiar with the data?
 - What are the problems to be solved and what could be the possible solution?
 - What skills are needed?
 - How will the model be evaluated?
- We must plan to allow the data analytics process to be robust, defining a workflow.

Data Analytics Cycle



Source: geeksforgeeks

Discovery (i)

1. Learn and investigate the problem
2. Develop context and understanding
3. Know the data sources needed and available for the project
4. Formulate the initial hypothesis that can later be tested with the data

Discovery (ii)

- Find an interesting topic
 - The project must be the answer to a **clear organizational need**
 - It should focus on the **big picture**, but also on the **specific objective**
 - Data Analytics is more than a science, it's an **art**, you have to like to walk the path, but the goal has to be something motivating
- Get and understand the data
 - There are many online data sources, e.g. at Kaggle, Google Cloud Public Datasets, Data.gov
 - Considering the type and format of the data, familiarization with the forms that the data can take, as well as its visualization and manipulation are very important factors.
 - Look for missing data, duplicate data, different spelling errors, or even data that doesn't make sense.

Discovery (iii)

- Points not to forget
 - Define/discover possible goals
 - Realize what data we have (discrete, continuous, categorical, boolean), get it from various sources
 - Understand the ranges of each variable
 - Know missing data
- Cross data in simple exploratory ways
- Not being your area **ASK / INVESTIGATE** to ensure that the end user is aligned with the objectives

Data preparation (i)

- Steps to explore, pre-process and prepare data before modeling and analysis
- Usually associated with a form of testing (analytical sandbox), executing, loading and transforming in order to place the data
- Data preparation tasks will likely be performed multiple times and not in a predefined order.
- Several tools can be used at this stage: Excel, Hadoop, Alpine Miner, Open Refine, etc.

Data preparation (ii)

- To perform any analytical activity on any data, the data needs to be in a structured format.
- The preparation (or **pre-processing**) step includes several tasks such as:
 - Data cleaning
 - Data organization.
 - Check data type compatibility
 - Checking for missing or outliers
 - Checking for discrepancies or errors
 - Creating substitution variables for categorical variables
 - Filtering of unnecessary variables
 - Creation of new variables (for example by composing others)

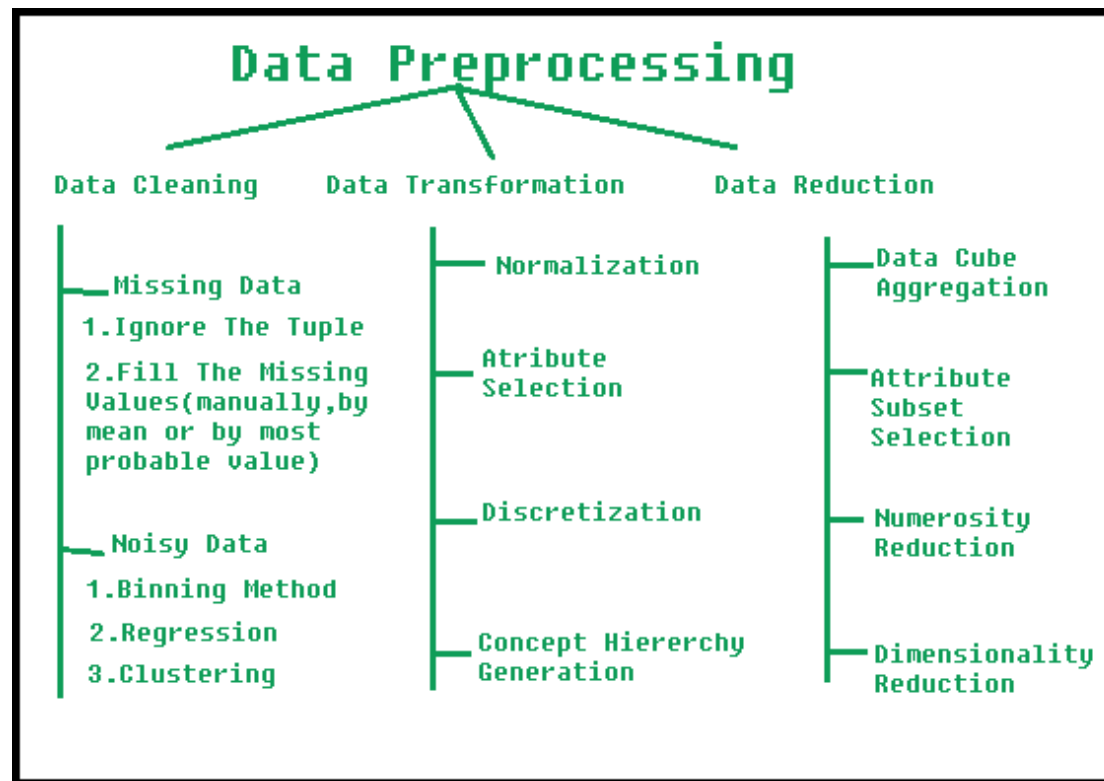
Data preparation – why?

- Data must be formatted for software
- Data must be formatted for a certain method
- Data is “dirty”:
 - **Incomplete**: attribute values are missing, attributes are missing or are aggregated and therefore difficult to distinguish
 - **Noisy**: containing errors or outliers (e.g. Salary=“-10”, Age=“222”)
 - **Inconsistent**: containing discrepancies in codes or names, for example:
Age=“22” BirthDate=“03/07/1976”
 - Rating “1,2,3” and rating “A, B, C”
 - Inconsistencies, e.g., Address: Universidade de Coimbra City: Lisboa

Data preparation (iii)

- **Data Discretization:** Part of data reduction, but with particular importance, especially for numerical data
- **Data Cleansing:** Fill in missing values, smooth out noisy data, identify and/or remove outliers, and resolve inconsistencies
- **Data Integration:** Integration of multiple data sources
- **Data Transformation:** Normalization and Aggregation
- **Data reduction:** Achieves reduced representation in volume but produces the same or similar analytical results

Data preparation (iv)



Fonte: geeksforgeeks

Data preparation steps

1. Data cleaning

- a. Missing data (skip rows or fill values with averages)
- b. Noisy data (implement outlier detectors)

2. Data transformation

- a. Normalization (scaling in a range, eg 0 to 1)
- b. Attribute selection (choose the ones that contain information)
- c. Discretization or binning (replacement of categories by values or ranges)
- d. Hierarchical replacement of concepts (e.g. city/country)

3. Data reduction

- a. data aggregation
- b. Selecting a subset of attributes
- c. Dimensionality reduction

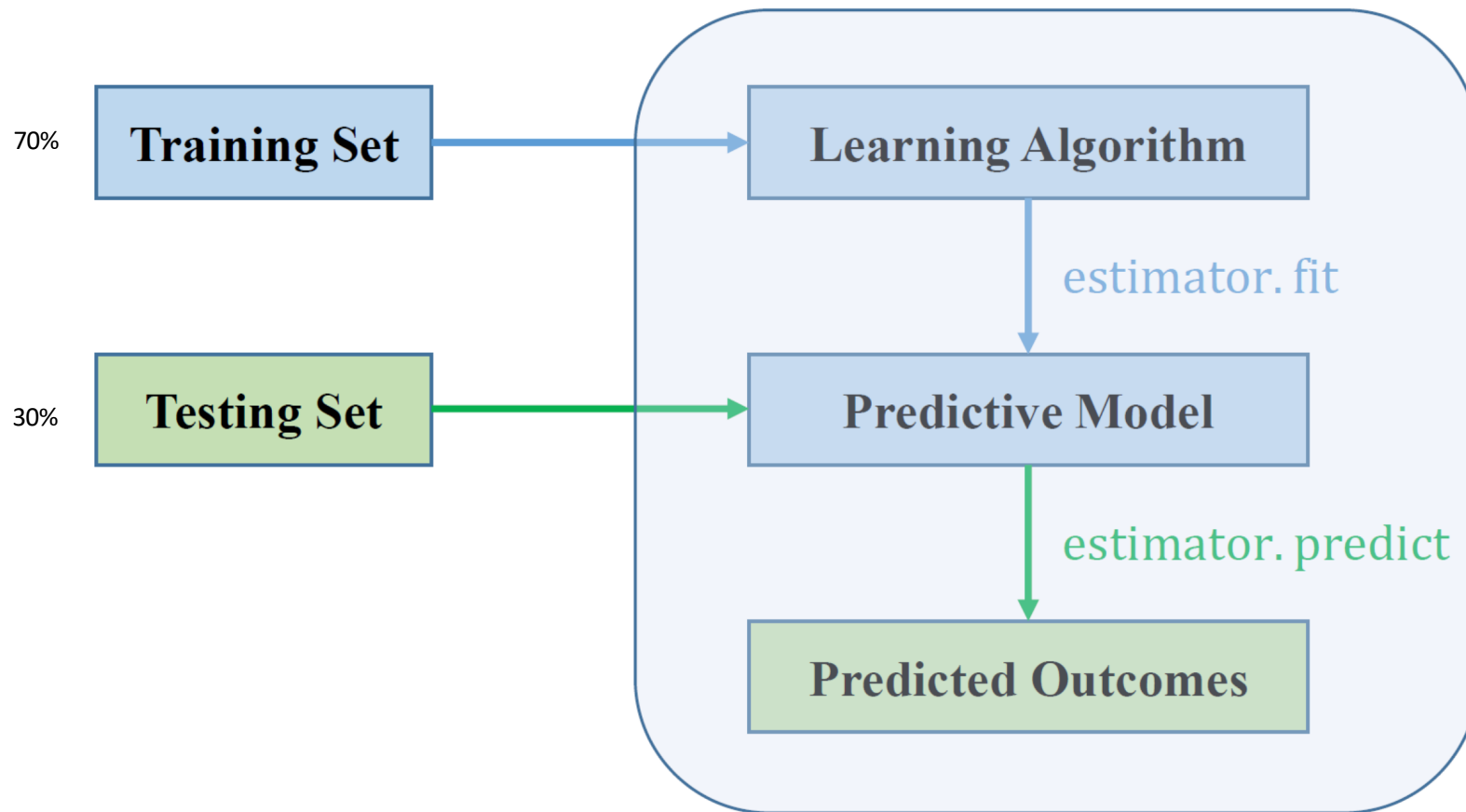
Data preparation example

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Model planning and construction

- Explore the data to learn the relationships between variables and subsequently select the most suitable key variables and models
- Create datasets for training, testing, and production purposes
- Consider whether existing tools are sufficient to run the models or if they need a more robust environment
- Several tools can be used at this stage:
 - Free or open source tools - R, Octave, WEKA, Python
 - Commercial tools - Matlab, STASTICA.

Training and testing sets



Fonte: towardsdatascience

Crossvalidation

- Alternative to the definition of training/test sets, which consists of performing k iterations and using the mean and/or standard deviation of the test results



Performance metrics – classification (i)

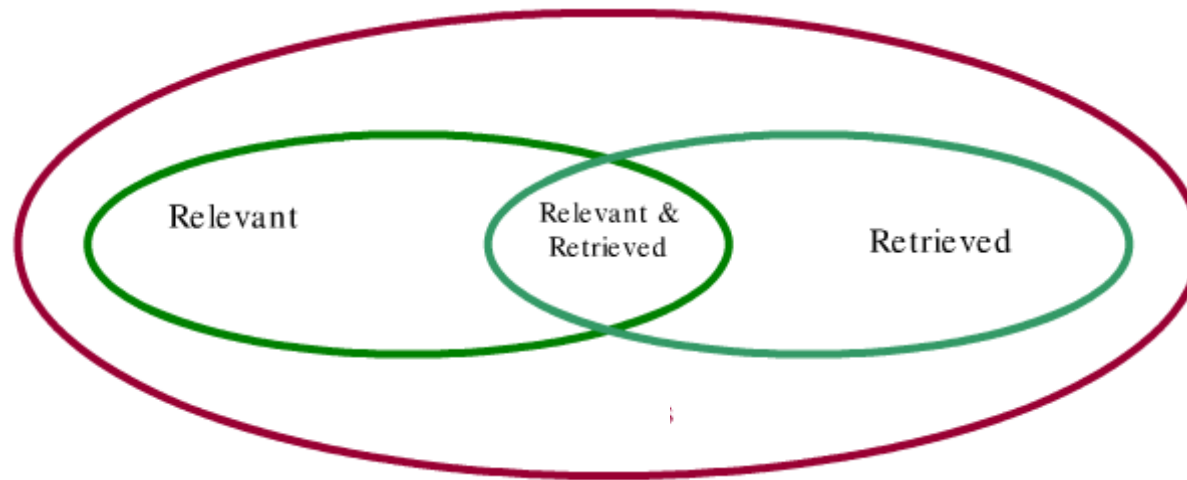
		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Contingency Matrix or Confusion Matrix

Performance metrics – classification (ii)

Metric	Formula	Evaluation Focus
Accuracy	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$	Overall effectiveness of a classifier
Error rate	$ERR = \frac{FP+FN}{TP+TN+FP+FN}$	Classification error
Precision	$PRC = \frac{TP}{TP+FP}$	Class agreement of the data labels with the positive labels given by the classifier
Sensitivity (Recall)	$SNS = \frac{TP}{TP+FN}$	Effectiveness of a classifier to identify positive labels
Specificity	$SPC = \frac{TN}{TN+FP}$	How effectively a classifier identifies negative labels
ROC	$ROC = \frac{\sqrt{SNS^2+SPC^2}}{\sqrt{2}}$	Combined metric based on the Receiver Operating Characteristic (ROC) space
F_1 score	$F_1 = 2 \frac{PRC \cdot SNS}{PRC + SNS}$	Combination of precision (PRC) and sensitivity (SNS) in a single metric
Geometric Mean	$GM = \sqrt{SNS \cdot SPC}$	Combination of sensitivity (SNS) and specificity (SPC) in a single metric

Performance metrics – classification (iii)



- **Precision:** % of examples chosen as positive (retrieved) that are in fact positive.
- **Recall:** % of positive examples that were effectively classified as positive.

Performance metrics – classification (iv)

- Example: Classification of email messages as SPAM (positive class) and non-SPAM (negative class).
- Let's consider 100 messages to classify, 80 legitimate and 20 SPAM. Let's also consider that our classifier finds 70 of the legitimate ones (non-SPAM) and 5 of the SPAM.
- Fill in the table:

		TRUE CLASS	
		P	N
PREDICTED CLASS	P	TP:_____	FP:_____
	N	FN:_____	TN:_____

Performance metrics – classification (v)

		TRUE CLASS	
		P	N
PREDICTED CLASS	P	TP: 5	FP: 10
	N	FN: 15	TN: 70

- Error: $25/100 = 25\%$
- Accuracy: $75/100 = 75\%$
- Recall: $5/20 = 25\%$
- Specificity: $70/80 = 88\%$
- Precision: $5/15 = 33\%$
- F1: 29%

Metric	Formula
Accuracy	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$
Error rate	$ERR = \frac{FP+FN}{TP+TN+FP+FN}$
Precision	$PRC = \frac{TP}{TP+FP}$
Sensitivity (Recall)	$SNS = \frac{TP}{TP+FN}$
Specificity	$SPC = \frac{TN}{TN+FP}$
ROC	$ROC = \frac{\sqrt{SNS^2 + SPC^2}}{\sqrt{2}}$
F ₁ score	$F_1 = 2 \frac{PRC \cdot SNS}{PRC + SNS}$

Performance metrics – regression

- Usually based on accounting for the error, that is, the difference between what was obtained and what was desired.
- Most common measures (y_i will be the prediction and $\hat{y}_i$ the desired):
- Mean Absolute Error

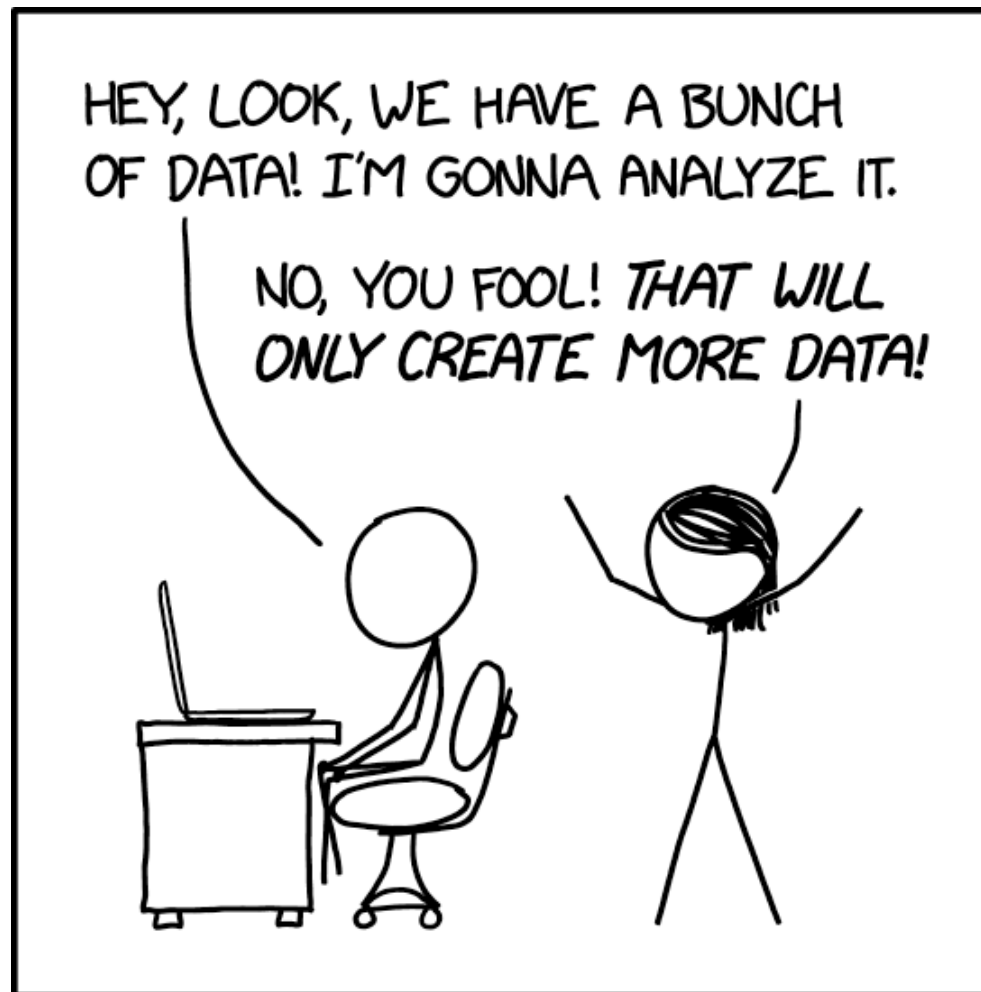
$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

- Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Communication and Operationalization

- Communicate project benefits more broadly and set up the pilot project to roll out the work in a controlled manner **before** scaling the work to the entire users enterprise
- This approach allows for knowledge about the performance and related constraints of the model in the small-scale production environment to make adjustments before deployment.
- Deliver final reports, briefings, codes
- Free or open source tools



<https://xkcd.com/2582/>